SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage : 20%

QUESTIONS: **Total Marks: 50**

Annexure A

Scenario-Based

Code of Conduct:

I (Deepak Kumar B/ 192424227) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A
Scenario-base d

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A. Explain why this issue occurs.
B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

A. Identify the most suitable algorithm for this scenario.
B. Justify your choice based on security strength and efficiency.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

A. Explain how the key exchange works in this scenario.
B. Identify a possible security threat and suggest a solution to mitigate it.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

A. Describe how RSA can be used for encryption and decryption in this scenario.
B. Explain how key distribution and management play a role in maintaining security.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices.
B. Recommend a suitable cryptographic approach and justify your answer.

Question 2: Choosing an Encryption Algorithm for a Banking System

Understanding of the Scenario

A bank encrypts customer transactions, so the data is highly sensitive and regulated. It needs strong confidentiality, resistance to brute-force attacks, and good throughput, since many transactions are processed continuously.

Key Issues to Consider

- Key length determines brute-force resistance.
- Known attacks and the algorithm's standing today.
- Speed and computational cost per transaction.
- Block size (all three use 64-bit blocks).

Comparison of the Algorithms

Feature	DES	3-DES	Blowfish
Key size	56 bits	112 or 168 bits (about 112 bits effective)	32 to 448 bits (variable)
Block size	64 bits	64 bits	64 bits
Rounds	16	48 (3 × 16)	16
Speed	Fast	Slow (3 passes of DES)	Very fast in software
Known weaknesses	Brute-forced in hours	Meet-in-the-middle, Sweet32	Sweet32 (64-bit block), weak keys in reduced rounds

A. Most Suitable Algorithm

Blowfish, used with a long key (256 bits or more).

B. Justification

Security strength:

- DES is ruled out immediately. Its 56-bit key can be brute-forced with modern hardware.
- 3-DES is much stronger, but its effective strength is only about 112 bits. It is also vulnerable to Sweet32 birthday attacks on its 64-bit block, and NIST has deprecated it.
- Blowfish allows keys up to 448 bits, and no practical attack on the full 16-round cipher is known.

Efficiency: 3-DES runs the DES operation three times per block, which makes it slow. Blowfish is much faster in software, which suits a high-volume transaction system.

Trade-off (high security plus moderate performance): Blowfish gives the best balance of the three. It has strong security through its large key and good speed through its simple Feistel structure.

Mitigations to state: Use a secure mode (CBC or CTR with unique IVs) and re-key regularly, because

the 64-bit block limits the amount of data safely encrypted under one key. Add a MAC for integrity.

Question 3: Key Exchange in an Unsecured Network

Understanding of the Scenario

Two users, Alice and Bob, must agree on a shared secret over a public network where attackers can listen to or alter messages. They cannot send the key directly, so they use Diffie-Hellman (DH).

A. How the Key Exchange Works

1. Alice and Bob publicly agree on a large prime **p** and a generator **g**.
2. Alice picks a private key **a** and sends **A = g^a mod p**.
3. Bob picks a private key **b** and sends **B = g^b mod p**.
4. Alice computes **K = B^a mod p**, and Bob computes **K = A^b mod p**.
5. Both get the same value, **K = g^(ab) mod p**, which becomes the shared session key.

Example with small numbers: p = 23, g = 5, a = 6, b = 15.

- A = 5^6 mod 23 = 8
- B = 5^15 mod 23 = 19
- Alice: 19^6 mod 23 = **2**
- Bob: 8^15 mod 23 = **2**

An eavesdropper sees p, g, A and B but cannot compute K without solving the **discrete logarithm problem**, which is computationally infeasible for large primes (2048 bits or more).

B. Security Threat and Mitigation

Threat: man-in-the-middle (MITM) attack. Basic DH has no authentication. An attacker, Mallory, intercepts A and B and substitutes her own values. She then shares one key with Alice and another with Bob, so she can read and modify all traffic while both users believe they are talking directly to each other.

Solutions:

6. **Authenticate the exchanged values** with digital signatures and certificates (PKI), as in TLS, or with pre-shared keys. Each party signs its DH value, so Mallory cannot substitute hers (the Station-to-Station protocol is one example).
7. **Use ephemeral DH (DHE/ECDHE)** for forward secrecy, so that compromising a long-term key does not expose past sessions.
8. **Use strong parameters:** a prime of at least 2048 bits, or elliptic-curve DH.
9. **Add key confirmation (MAC)** to verify that both sides derived the same key.

Conclusion

DH lets two parties create a shared secret over an open network, but it must be combined with authentication to resist MITM attacks.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand